

MAT 524: Functions of a Real Variable II

Homework I

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[[R & F: Sec. 4.4, Pr. 28]]:

Statement

Let f be integrable over E and let C be a measurable subset of E . Then

$$\int_C f = \int_E f \chi_C.$$

Proof

Since f is integrable over E so is $|f|$, by definition. Clearly, $|f \chi_C| \leq |f|$ on E , so by the Integral Comparison Test (Proposition 16), it follows that $f \chi_C$ is integrable over E . Thus, we can write

$$\int_E f \chi_C = \int_E [f \chi_C]^+ - \int_E [f \chi_C]^- \tag{1}$$

$$= \int_E \max\{f \chi_C, 0\} - \int_E \max\{-f \chi_C, 0\} \tag{2}$$

$$= \int_E \max\{f, 0\} \chi_C - \int_E \max\{-f, 0\} \chi_C \tag{3}$$

$$= \int_E f^+ \chi_C - \int_E f^- \chi_C. \tag{4}$$

The functions f^+ and f^- are both non-negative, so using the theory from Section 4.3 of Royden and Fitzpatrick, we see that

$$\int_E f^+ \chi_C - \int_E f^- \chi_C = \int_C f^+ - \int_C f^- \tag{5}$$

$$= \int_C f. \tag{6}$$

Therefore, $\int_C f = \int_E f \chi_C$, as required. ■



[[R & F: Sec. 4.4, Pr. 29]]:

Problem

For a measurable function f on $[1, \infty)$ which is bounded on bounded sets, define $a_n = \int_n^{n+1} f$ for each natural number n . **(a)** Is it true that f is integrable over $[1, \infty)$ if and only if the series $\sum_{n=1}^{\infty} a_n$ converges? **(b)** Is it true that f is integrable over $[1, \infty)$ if and only if the series $\sum_{n=1}^{\infty} a_n$ converges absolutely?

Solution

(a) If f is integrable over $[1, \infty)$ then, by definition, $\int_{[1, \infty)} |f|$ converges. But

$$\int_{[1, \infty)} |f| = \int_1^2 |f| + \int_2^3 |f| + \int_3^4 |f| + \dots \quad (7)$$

$$\geq \left| \int_1^2 f \right| + \left| \int_2^3 f \right| + \left| \int_3^4 f \right| + \dots \quad (8)$$

$$= \sum_{n=1}^{\infty} |a_n|, \quad (9)$$

implying that $\sum_{n=1}^{\infty} a_n$ converges absolutely. Thus, $\sum_{n=1}^{\infty} a_n$ converges.

As for the converse statement, it is not necessarily true. Consider the following counter-example. Let $f(x) = \pi \cos(\pi x)$. Then $a_n = \int_n^{n+1} \pi \cos(\pi x) dx = \sin((n+1)\pi) - \sin(n\pi) = 0$ for all n . Clearly $\sum_{n=1}^{\infty} a_n$ converges, but $\int_{[1, \infty)} \pi \cos(\pi x) dx = \lim_{x \rightarrow \infty} \sin(\pi x) - \sin(\pi)$ diverges.

(b) We already showed in the previous part that if f is integrable over $[1, \infty)$ then $\sum_{n=1}^{\infty} a_n$ converges absolutely.

The converse statement also doesn't hold in this case. The same counter-example can be used here, since $\sum_{n=1}^{\infty} a_n$ (with a_n defined as in part (a)) also converges absolutely.

■

